SET UP MONITORING DASHBOARDS Lab Documentation Report

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Tools: Kibana | Elastic Security | Grafana

1. Objective

The objective of this activity was to create monitoring dashboards for security event visualization and analysis. Dashboards were configured to display the most active source IP addresses and the frequency of critical Windows security events. These visualizations assist SOC analysts in identifying suspicious activity and monitoring security trends.

2. Dashboard Configuration

The monitoring dashboards were created using Kibana and Elastic Security.

Dashboard 1

Top 10 Source IPs Generating Alerts Purpose:

- Identify systems generating the highest number of security alerts.
- Detect suspicious hosts and potential attack sources.
- Support threat hunting and incident investigations.

Data Source

Security event logs collected from:

- Windows VM
- Wazuh Agent
- Elastic SIEM

Example Results Source IP Alert Count

192.168.56.101 45

192.168.56.104 18

192.168.56.103 10

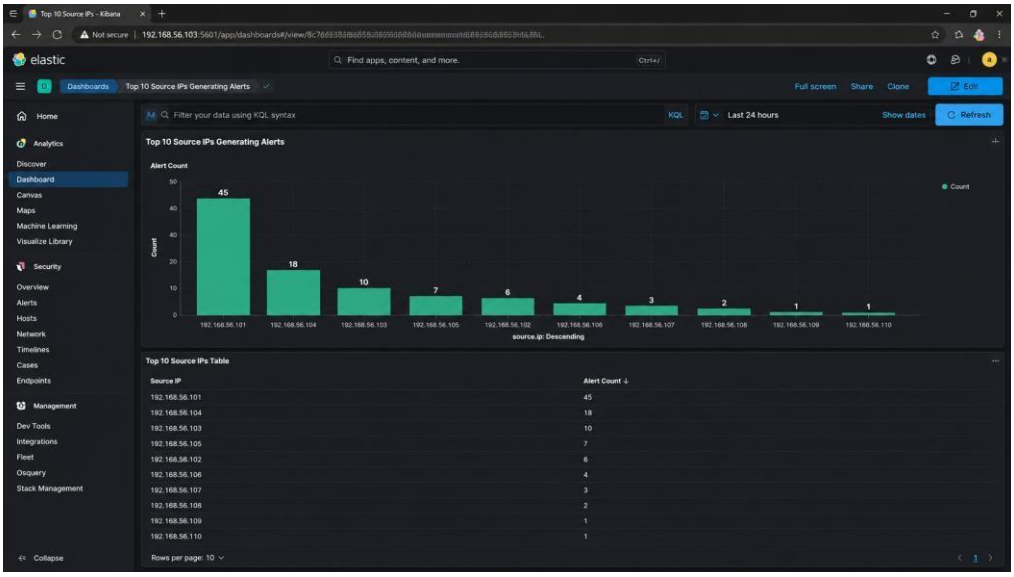

Figure 1: Dashboard visualization displaying the top source IP addresses responsible for security alerts.

3. Critical Event ID Monitoring

A second dashboard was created to monitor the frequency of critical Windows Event IDs.

Event IDs Monitored Event ID Description

4625	Failed Login Attempt
4624	Successful Login
7045	Service Installation
4688	Process Creation

Purpose

- Detect brute-force attacks.
- Monitor account activity.
- Track service installations.
- Identify suspicious process execution.

Example Event Frequency Event ID Count

4625	50
4624	30
7045	5
4688	15

Figure 2: Dashboard showing frequency analysis of critical Windows security events.

4. Sigma Rule Integration

Pre-built Sigma detection rules were used to improve monitoring coverage and identify suspicious activities.

Example Detection Rules

- Brute Force Login Detection
- Suspicious Service Installation
- PowerShell Execution Monitoring
- Account Enumeration Detection

These rules enhanced visibility into security events and supported faster incident detection.

5. Findings

The monitoring dashboards provided centralized visibility into security events generated within the lab environment. The visualizations enabled rapid identification of high-risk hosts and frequently occurring security events. Event ID 4625 was observed as the most common alert due to brute-force login simulations originating from IP address 192.168.56.101.

6. Conclusion

The dashboards successfully visualized security events collected by Elastic Security and Wazuh. The Top Source IP dashboard identified the most active systems, while the Event Frequency dashboard provided insight into critical Windows security events. These dashboards improved monitoring effectiveness and supported SOC analysis activities.